

Previews

hPSCs-derived Schwann cells serve for disease modeling and drug discovery

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In this issue, Majd et al.¹ derive Schwann cells from human pluripotent stem cells (hPSCs), which can be used to study Schwann cell development and physiology and model diabetic neuropathy. hPSC-derived Schwann cells possess the molecular features of primary Schwann cells and are capable of myelination *in vitro* and *in vivo*.

Schwann cells are a vital component of the peripheral nervous system (PNS), where they play a crucial role in the development, maintenance, and regeneration of peripheral nerves. These cells are responsible for myelinating the axons of motor and sensory neurons in the PNS, providing electrical insulation that facilitates efficient signal transmission. Schwann cells also aid in axonal guidance, trophic support, and the regulation of immune responses in the PNS.²

However, Schwann cells can be affected in various disorders, resulting in peripheral neuropathies, a class of diseases that affect the PNS.³ Peripheral neuropathies can be classified as demyelinating or axonal depending on the primary site of damage. In demyelinating neuropathies, the myelin sheath is damaged, resulting in slowed or blocked nerve impulses, while in axonal neuropathies, the axon is damaged, resulting in the loss of nerve function.

One of the most common Schwann cell-based peripheral neuropathies is diabetic peripheral neuropathy (DPN), which affects up to 50% of individuals with diabetes. DPN is characterized by nerve damage resulting from high blood glucose levels, which can damage the blood vessels supplying the nerves or directly damage the Schwann cells. DPN usually presents as a loss of sensation in the feet, which can progress to involve the hands and other parts of the body. DPN can also cause pain, weakness, and muscle wasting in affected areas.⁴ The treatment of DPN is often individualized, and the choice of treatment depends on the severity of the symptoms, the patient's medical history,

and other factors. Most of the current treatments are linked to maintaining good blood sugar control and managing symptoms with pain relievers, antidepressants, and anticonvulsants.^{4,5}

Developing *in vitro* models for DPN is crucial for several reasons. First, DPN is a complex and multifactorial disease, making it difficult to study *in vivo* and to cure. *In vitro* models allow for more controlled experiments and a better understanding of the cellular and molecular mechanisms underlying DPN. Next, *in vitro* models can help to test drugs and other therapies in a controlled environment, which can provide insights into their effectiveness and potential side effects before they are tested in humans.⁶ This can help reduce the number of animal and human experiments needed, making drug development more efficient and ethical. Finally, *in vitro* models of DPN can be used to study the effects of environmental factors, such as high glucose levels, on Schwann cells and other cells involved in peripheral neuropathies.

In their recent manuscript, Majd et al.¹ have developed an approach to derive bio-equivalent Schwann cells from human pluripotent stem cells (hPSCs), which can be used to study Schwann cell development, physiology, and model DPN. These hPSC-derived Schwann cells possess the molecular features of primary Schwann cells and are capable of myelination in both *in vitro* and *in vivo* settings. Importantly, Majd et al. used single-cell transcriptomics datasets⁷ of adult human Schwann cells as a benchmarking reference to validate the identity of the *in vitro* hPSC-derived Schwann cells. They also found that the cultured mature hPSC-

derived Schwann cells matched the gene expression pattern of adult Schwann cells. This validation strengthens confidence in the protocol to a degree matched by only a few other recently developed *in vitro* generated Schwann cell models.⁸

In the following step, Majd et al. took advantage of their original human Schwann cell *in vitro* system to model DPN and demonstrated that high glucose levels can result in Schwann cell death and myelin damage in a controlled environment. However, they also discovered that bupropion, an FDA-approved antidepressant drug,⁹ could counteract this glucotoxicity and prevent sensory dysfunction, Schwann cell death, and myelin damage in hyperglycemic mice. This is of significant value, as approved drugs have already been evaluated for their safety and tolerability in humans. Therefore, repurposing these drugs for new diseases can minimize the risks associated with drug development.¹⁰ The safety profile of these drugs is already established, and knowing this can expedite the regulatory approval process, especially in the case of DPN.

Majd et al. found that bupropion's ability to counteract glucotoxicity correlated with a decrease in intracellular glucose and sucrose levels. Bupropion treatment resulted in reduced cellular lactate concentration and its membrane transporters, which might be potentially important given the key role of lactate as a fuel in neuronal metabolism. The drug also reversed the downregulation of genes associated with mitochondrial function and oxidative phosphorylation in Schwann cells exposed to high glucose. These results suggest that bupropion ameliorates the negative effects



of high glucose on Schwann cell metabolism and mitochondrial function, which in turn prevents myelin damage and Schwann cell death.

Majd et al.'s retrospective analysis of health records revealed that bupropion treatment is associated with a lower incidence of neuropathy among diabetic patients. This finding suggests that bupropion may have therapeutic potential in preventing or treating DPN in humans. In conclusion, bupropion could be a promising therapeutic option for preventing or treating DPN.

The initial *in vitro* screening of compounds presented by Majd and coauthors has identified promising hits beyond bupropion, but additional investigations are required to confirm their effectiveness and to understand their mechanisms of action. Therefore, future studies are needed to continue the exploration of these compounds' therapeutic potential, which may lead to the development of new treatments for DPN and related diseases.

All these advancements aid the prospective research endeavors in the areas of Schwann cell-mediated neuronal maturation and myelination, as well as the modeling of PNS pathologies such as DPN. Developing an advanced *in vitro* Schwann cell model for combinatorial screening of compounds that modulate peripheral myelination or alleviate the

symptoms of disease at the cellular level could be the next step in this direction. Furthermore, autologous Schwann cells are currently under trial for their use in regenerative medicine applications aimed at treating both PNS and CNS disorders. The transplantation data, which exhibit strong engraftment of hPSC-Schwann cells in a model of traumatic nerve injury, further support their potential therapeutic role. Altogether, the *in vitro* Schwann cell derivation holds a wide range of potential biomedical applications, including modeling DPN.

DECLARATION OF INTERESTS

The authors declare no conflict of interests.

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